



Batterjee Medical College
For Science and Technology

An approach to a case of cough

Chronic Obstructive Pulmonary Disease (COPD)

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Learning Objectives

Knowledge:

- Define Chronic Obstructive Pulmonary Disease (COPD), chronic bronchitis, and emphysema, and identify global prevalence and epidemiology.
- Identify exogenous risk factors and endogenous predisposing conditions
- Explain the mechanisms of chronic airway inflammation, mucus hypersecretion, peribronchiolar fibrosis, alveolar destruction, elastic recoil loss, air trapping, and V/Q mismatching.
- Describe typical clinical features including chronic cough, dyspnea, wheezing, chest hyperinflation, pursed-lip breathing, and cyanosis.
- Explain spirometric grading (GOLD 1–4 based on $FEV_1\%$) and clinical risk categorization (GOLD Groups A, B, and E using mMRC/CAT scores and exacerbation history).
- Outline non-pharmacological interventions (smoking cessation, vaccines, pulmonary rehab), stable COPD maintenance regimens (SABA, LABA, LAMA, ICS), and acute exacerbation strategies (oxygen targets, systemic steroids, bronchodilators, antibiotics, NIPPV).

Skills:

- Analyze patient histories and physical findings to differentiate between COPD clinical phenotypes and other obstructive airway diseases.
- Interpret spirometry parameters, arterial blood gas values, and imaging studies .
- Calculate mMRC scores and apply GOLD criteria to categorize disease severity and exacerbation risk.
- Formulate evidence-based treatment plans for stable COPD and acute exacerbations (AECOPD), including identifying indications for non-invasive positive pressure ventilation.

Values:

- Demonstrate a professional attitude through strict adherence to training site rules, safety protocols, and attendance policies.
 - Value patient education regarding smoking cessation, proper inhaler technique, and long-term compliance.
- Empathize with the physical and psychological burden of chronic dyspnea and advanced disease complications in COPD patients.



- Definition
- Etiology
- Pathophysiology
- Clinical Features
- Diagnostics
- Classification
- Management
- Subtypes
- Acute COPD exacerbation

Introduction

Obstructive lung diseases:

- a lung condition characterized by persistent respiratory symptoms (cough, dyspnea) and airflow limitation which is caused by a combination of small airway obstruction and parenchymal destruction.

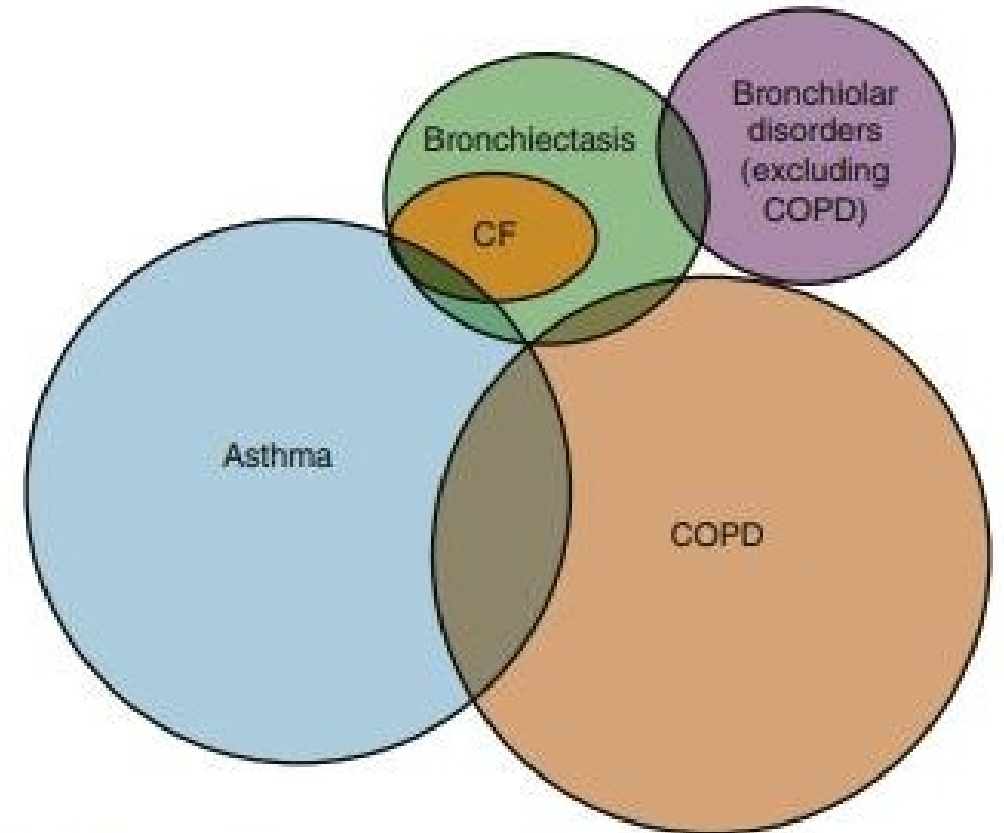


Fig. 16.1 Classification of obstructive lung diseases. Although most patients with chronic obstructive pulmonary disease (COPD) have small airways disease, the bronchiolar disorders do not overlap with COPD. CF, Cystic fibrosis.

Definitions

- **Chronic bronchitis:** productive cough for at least 3 months per year for 2 consecutive years that cannot be explained by an alternative diagnosis.
- **Emphysema:** permanent dilatation of pulmonary air spaces distal to the terminal bronchioles that is caused by the destruction of the alveolar walls and pulmonary capillaries required for gas exchange

Global Initiative for Chronic
Obstructive
Lung
Disease



Etiology

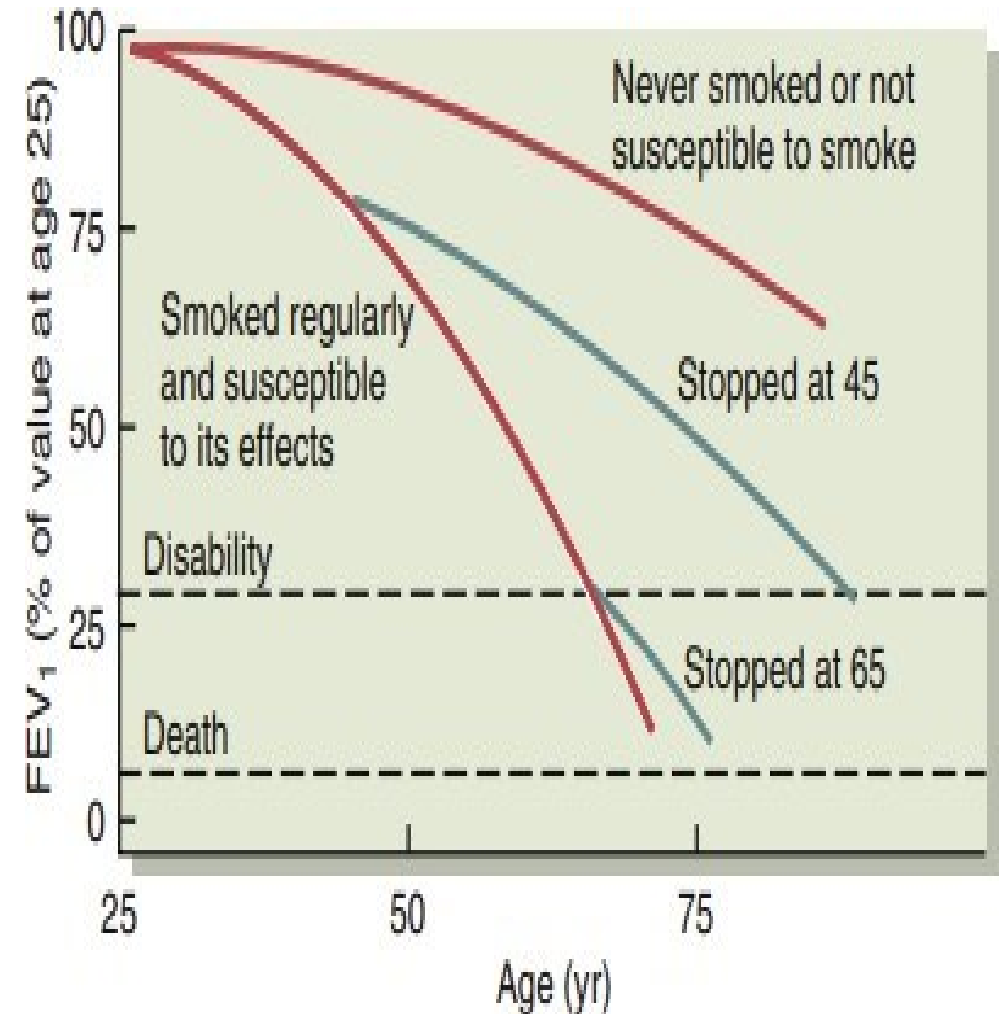
Risk factors & causes:

Exogenous factors:

- Tobacco smoke (indicated in almost 90% of COPD cases)
- Exposure to air pollution or fine dusts

Endogenous factors

- α 1-Antitrypsin deficiency
- Lung growth and development abnormalities
 - Recurrent pulmonary infections and tuberculosis
 - Premature birth
- Airway hyperresponsiveness
- Antibody deficiency syndrome (e.g., IgA deficiency)
- Primary ciliary dyskinesia (e.g., Kartagener syndrome)



Pathophysiology

COPD is characterized by chronic **airway** inflammation and tissue destruction.

Chronic inflammation:

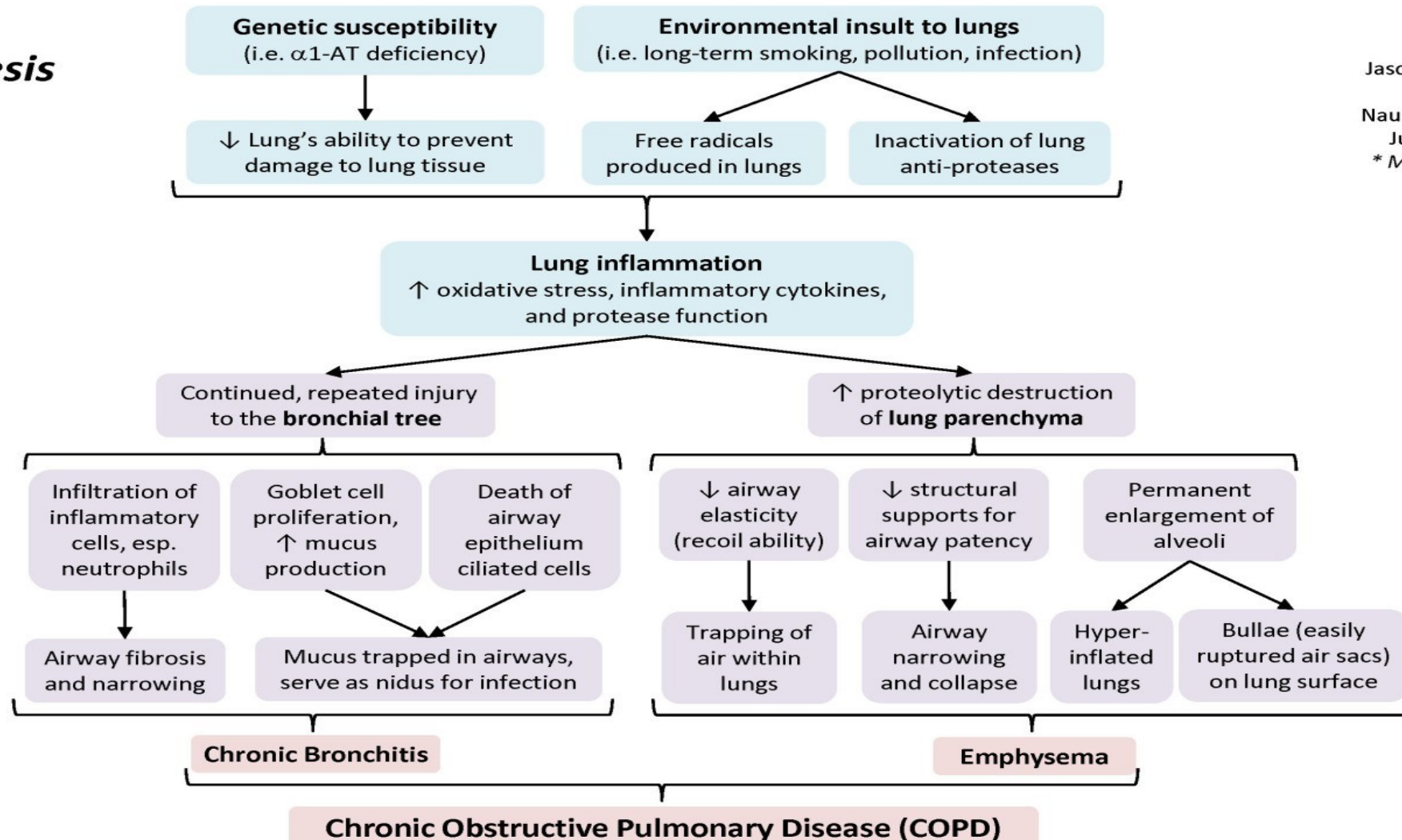
- Increased number of neutrophils → release of cytokines.
- Overproduction of growth factor → narrowing of airway.
- Promotion of goblet cell proliferation and hypertrophy, mucus hypersecretion → chronic productive cough
- hypoxic vasoconstriction in pulmonary vasculature → pulmonary hypertension

Tissue destruction:

- Nicotine use inactivates protease inhibitors (especially α_1 -antitrypsin) → imbalance of protease and antiprotease → loss of elastic tissue, which causes:
 - Enlargement of airspaces → $\downarrow$ DL_{CO} → hypoxemia and hypercapnia

Pathophysiology

COPD: *Pathogenesis*



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GOLD classification

GOLD spirometric grades ^[1]

Grade	Postbronchodilator FEV ₁ % of the predicted value
GOLD 1: mild	≥ 80%
GOLD 2: moderate	50–79%
GOLD 3: severe	30–49%
GOLD 4: very severe	< 30%

GOLD groups

GOLD group assessment

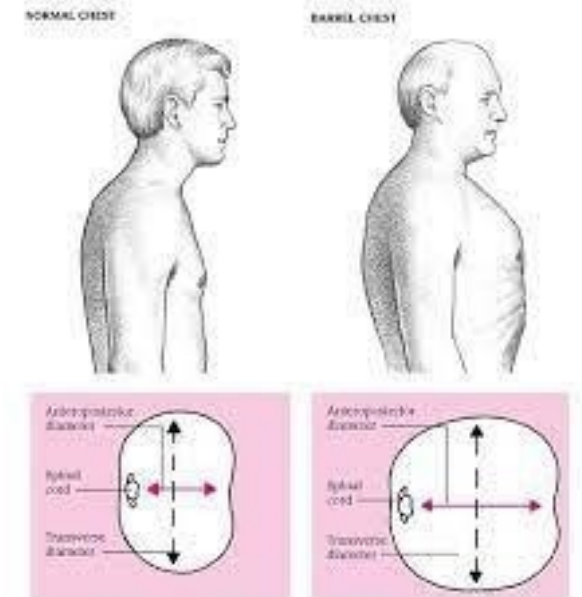
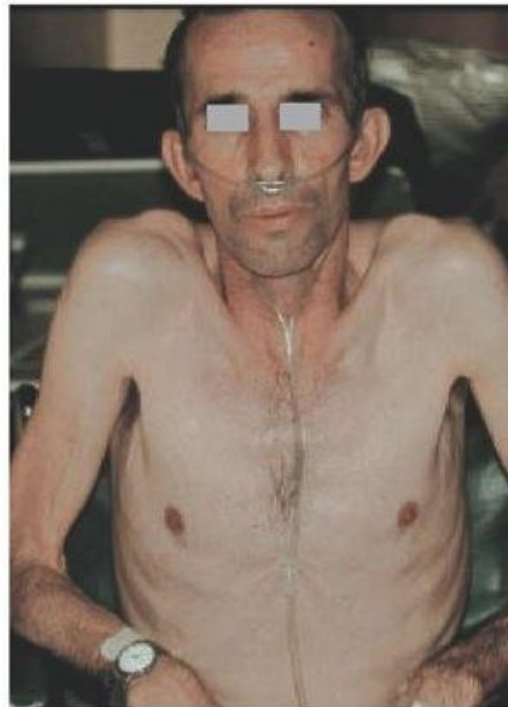
Group	Exacerbations in the past year	Severity of symptoms	
		mMRC dyspnea scale	CAT score
GOLD group A	•0	•0–1	•0–9
GOLD group B	•Or 1 not leading to hospital admission	•≥ 2	•≥ 10
GOLD group E	•≥ 2 •Or ≥ 1 leading to hospital admission	•Any severity	

Clinical Features

Symptoms & Signs:

- Symptoms are minimal or nonspecific until the disease reaches an advanced stage.
- Chronic cough & prolonged expiration
- Dyspnea
- Wheezing
- Chest wall hyperinflation
- Distant breath and heart sounds
- Clubbing is not seen in COPD
- Use of respiratory accessory muscles
- cyanosis “blue bloater”
- pursed-lip breathing “pink puffer”

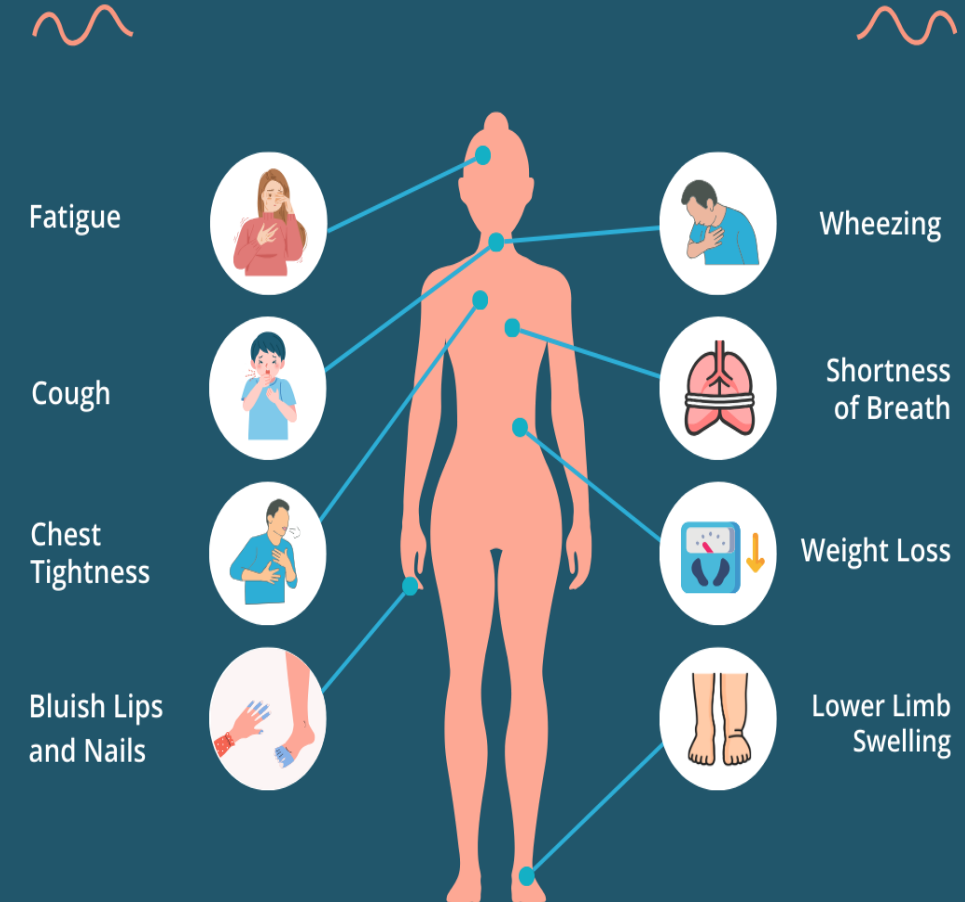
Clinical Features



Features of advanced COPD

- Congested neck veins
- **Barrel chest:** is most commonly seen in individuals with emphysema.
- Asynchronous movement of the chest and abdomen during respiration
- Use of accessory respiratory muscles due to diaphragmatic dysfunction
- Hyperresonant lungs, reduced diaphragmatic excursion, and relative cardiac dullness on percussion
- Decreased breath sounds on auscultation: “silent lung”
- Peripheral edema
- Right ventricular hypertrophy and cor pulmonale
- Hepatomegaly (congested liver)
- Often weight loss and cachexia
- Secondary polycythemia
- Confusion: due to hypoxemia and hypercapnia

How COPD Affects the Body



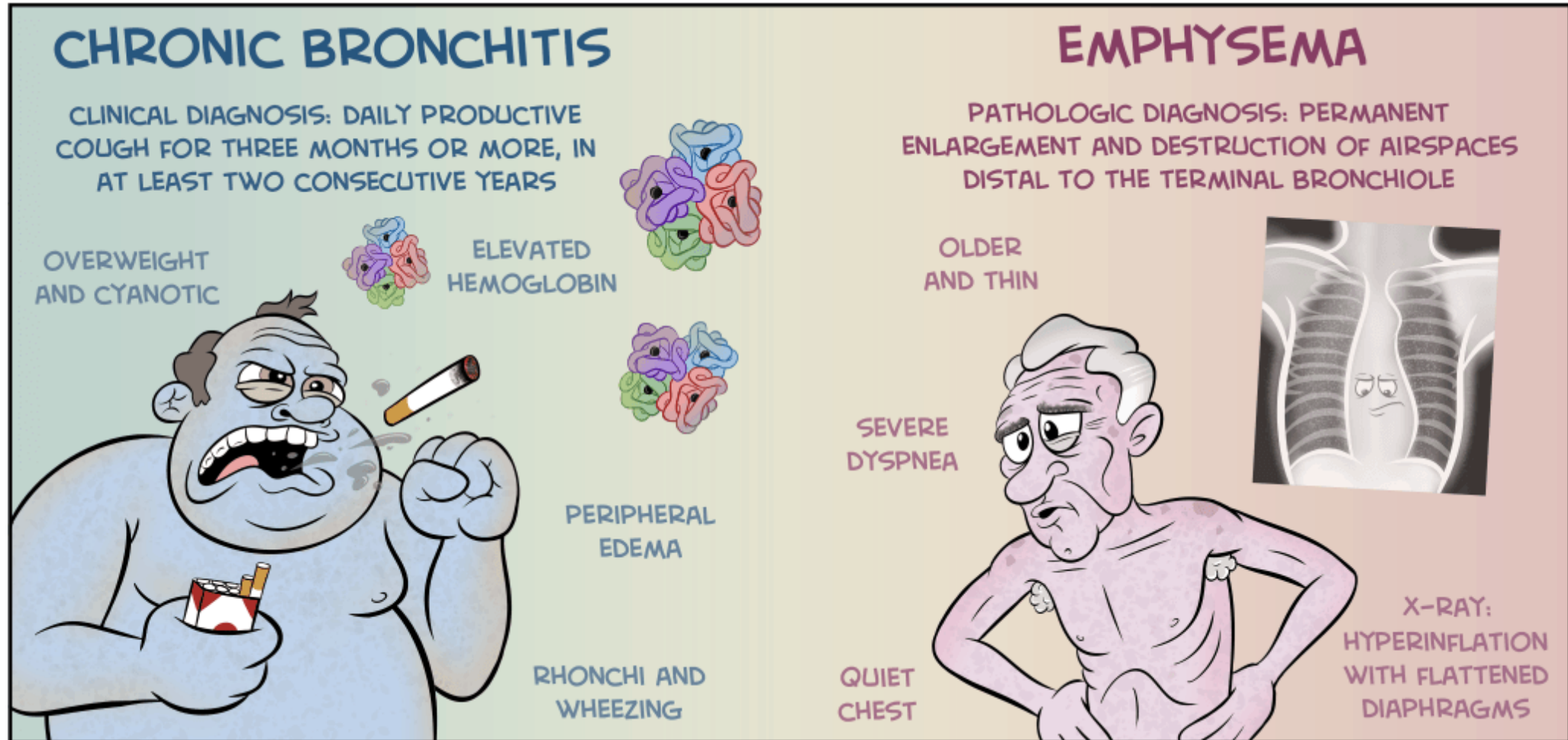
Types COPD

- Usually difficult to differentiate between the two types, & combination of them Usually exists

Pink puffer vs. blue bloater

	Pink Puffer	Blue Bloater
Main pathomechanism	• Emphysema	• Chronic bronchitis
Clinical features	• Noncyanotic • Cachectic • Pursed-lip breathing • Mild cough	• Productive cough • Overweight • Peripheral edema
PaO₂	• Slightly reduced	• Markedly reduced
PaCO₂	• Normal (possibly in late hypercapnia)	• Increased (early hypercapnia)

COPD comes in two flavors



Diagnostics

History of smoking or exposure to noxious gaseous
Clinical examination
Investigations

Diagnostics

CXR

- ↑ anteroposterior diameter
- Flattened diaphragm
- Transverse Ribs
- Long narrow heart shadow
- widened intercostal spaces
- Hyperlucency of lung tissue



E-Fig. 16.6 Anterior-posterior (*left*) and lateral (*right*) chest radiographs show changes caused by severe chronic obstructive pulmonary disease, including hyperinflation, increased anterior-posterior diameter, flattened diaphragms, and enlarged pulmonary arteries.

Diagnostics

Chest CT

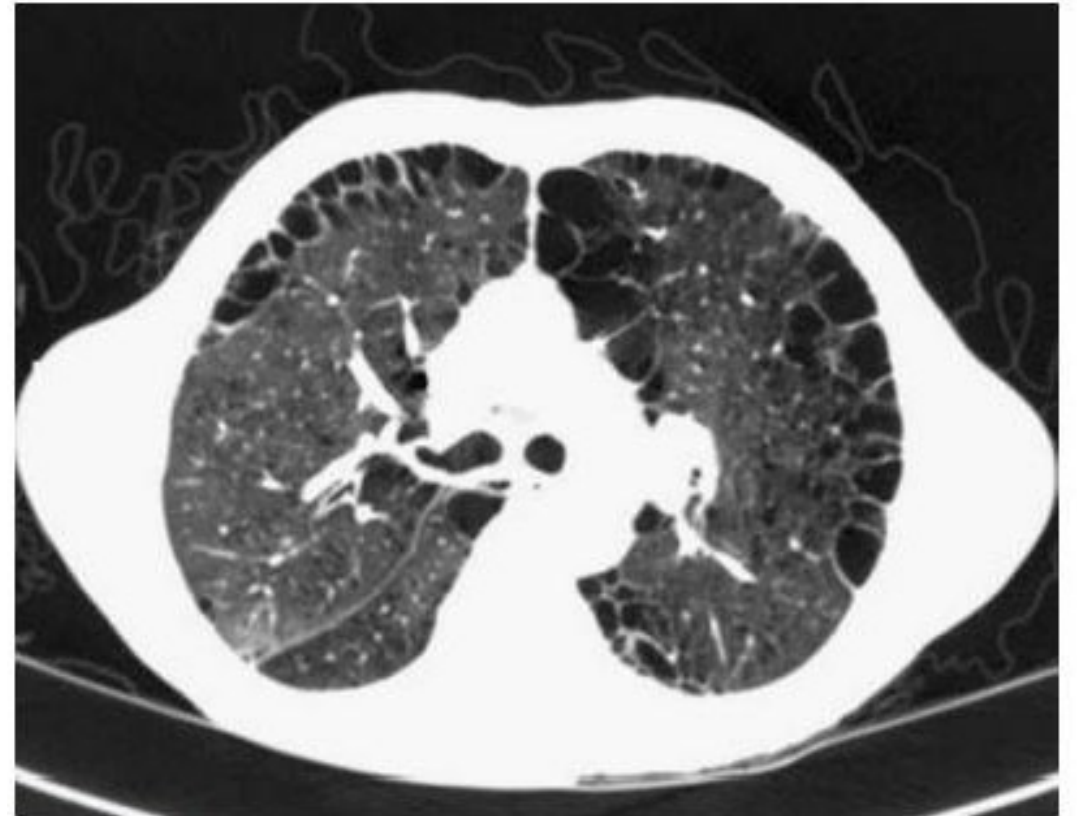
- Similar to CXR

Serum AAT

- Screen all patients (especially if young)

ABG

- chronic PCO₂ retention
- ↓ PO₂



E-Fig. 16.7 Computed tomographic scan shows paraseptal distribution of emphysema.

Diagnostics

PFT (Spirometry)

- ↓ FEV₁/FVC ratio (< 70%)
- ↓ FEV₁
- ↑ Total lung capacity (TLC)
- ↑ Functional residual capacity (FRC)
- ↑ Residual volume (RV)

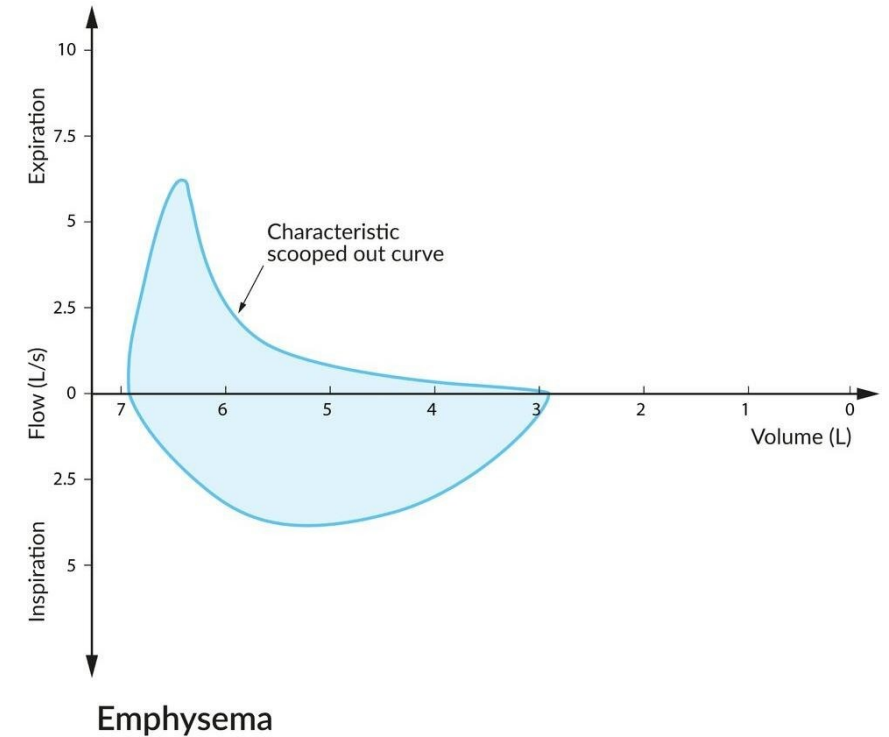
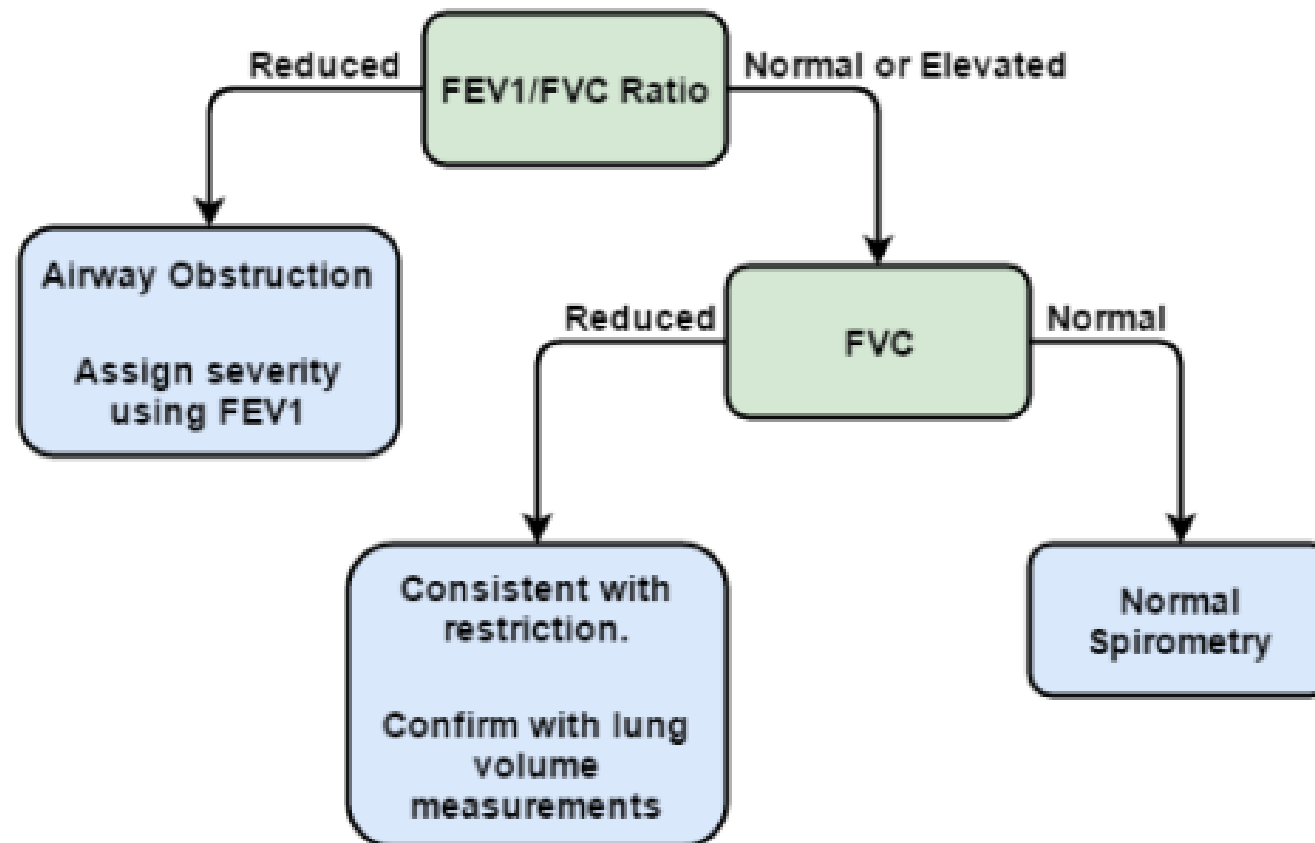


TABLE 16.2 GOLD Criteria

GOLD 1/mild COPD	FEV ₁ ≥80% predicted
GOLD 2/moderate COPD	FEV ₁ ≥50% but less than 80% predicted
GOLD 3/severe COPD	FEV ₁ ≥30% but less than 50% predicted
GOLD 4/very severe COPD	FEV ₁ <30% predicted

Diagnostics

PFT (Spirometry)



Management

- **Assess for both:**

- Symptom severity and history of exacerbations to determine the GOLD group
- Comorbidities that may influence treatment, e.g., asthma

- Provide nonpharmacological management of COPD for all patients.

- Initiate pharmacological treatment of COPD based on GOLD group.

- At follow-up:

- Review inhaler technique.
- Adjust medication based on the adequacy of symptom control.
- Repeat spirometry annually to track any decline in FEV_1 .

- In patients with advanced disease, consider:

- Long-term oxygen therapy
- Interventional treatment

Management

Nonpharmacological

- Smoking cessation
- Immunizations (influenza & pneumococcus)

Pharmacological

- Oxygen therapy ($SpO_2 \leq 88\%$ rest)
- Inhaled Muscarinic antagonists (main Line)
- Inhaled corticosteroids
- Inhaled β_2 -agonists
- Pulmonary rehab
- **Mucolytics** (e.g., N-acetylcysteine, erdosteine)
 - Liquefy mucus by reducing the disulfide bonds of mucoproteins
 - Can be useful in reducing exacerbations in certain patients
 - Additional (surgical, Abx, PDE-4 inhibitors, opiates, thiopheline)

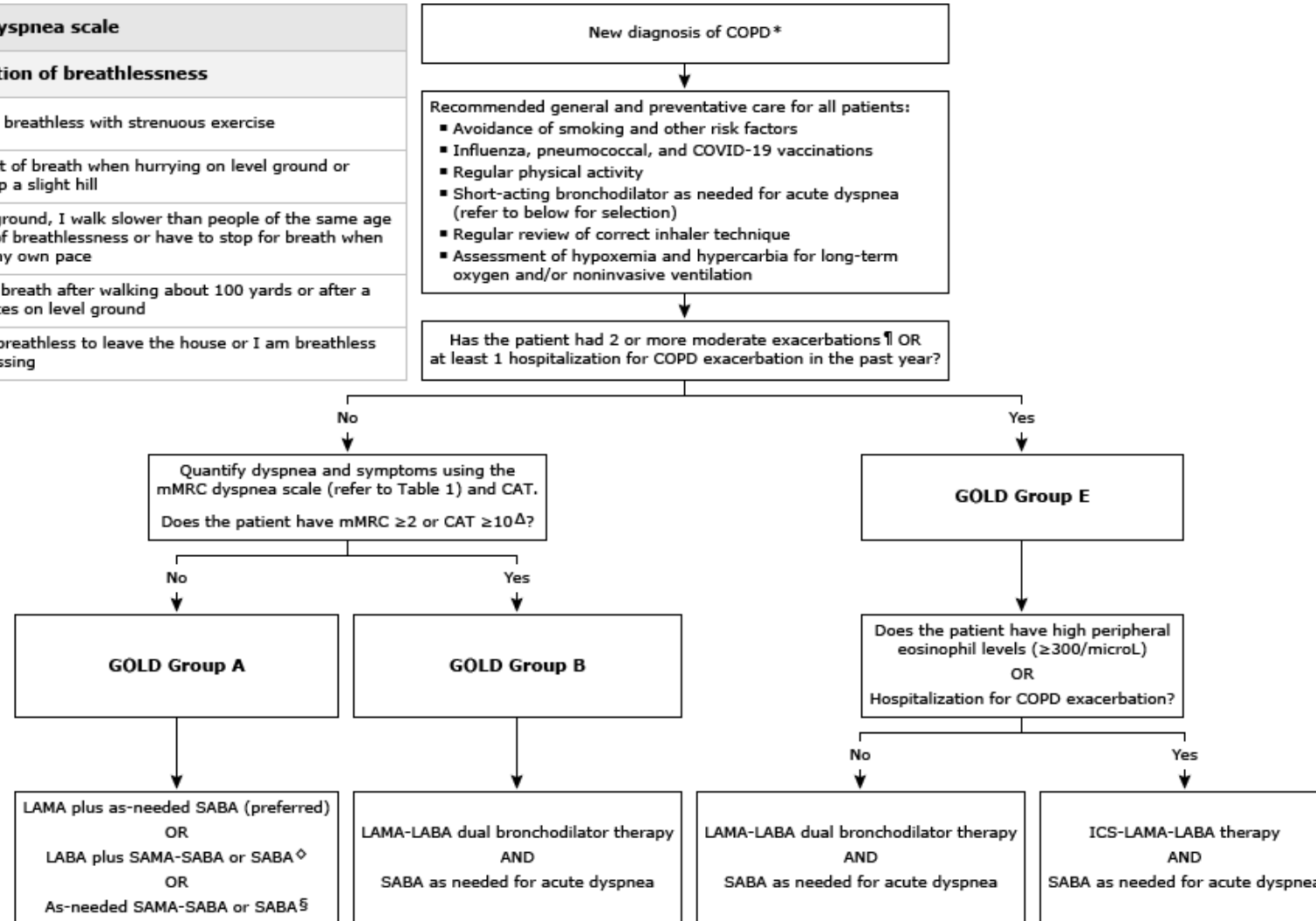
	LABA	LAMA	Inhaled steroids
12 hours	Salmeterol	Acclidinium	Beclomethasone
	Formoterol	Darotropium	Budesonide
24 hours			Fluticasone propionate
	Indacaterol	Tiotropium	Ciclesonide
	Vilanterol	Glycopyrronium	Mometasone
	Olodaterol	Umeclidinium	Fluticasone furoate
	Abediterol		
	Carmoterol		

Abbreviations: COPD, chronic obstructive pulmonary disease; LABA, long-acting beta agonist; LAMA, long-acting muscarinic antagonist.

Management

Table 1: mMRC dyspnea scale

Grade	Description of breathlessness
0	I only get breathless with strenuous exercise
1	I get short of breath when hurrying on level ground or walking up a slight hill
2	On level ground, I walk slower than people of the same age because of breathlessness or have to stop for breath when walking my own pace
3	I stop for breath after walking about 100 yards or after a few minutes on level ground
4	I am too breathless to leave the house or I am breathless when dressing



Acute COPD Exacerbation

Definition:

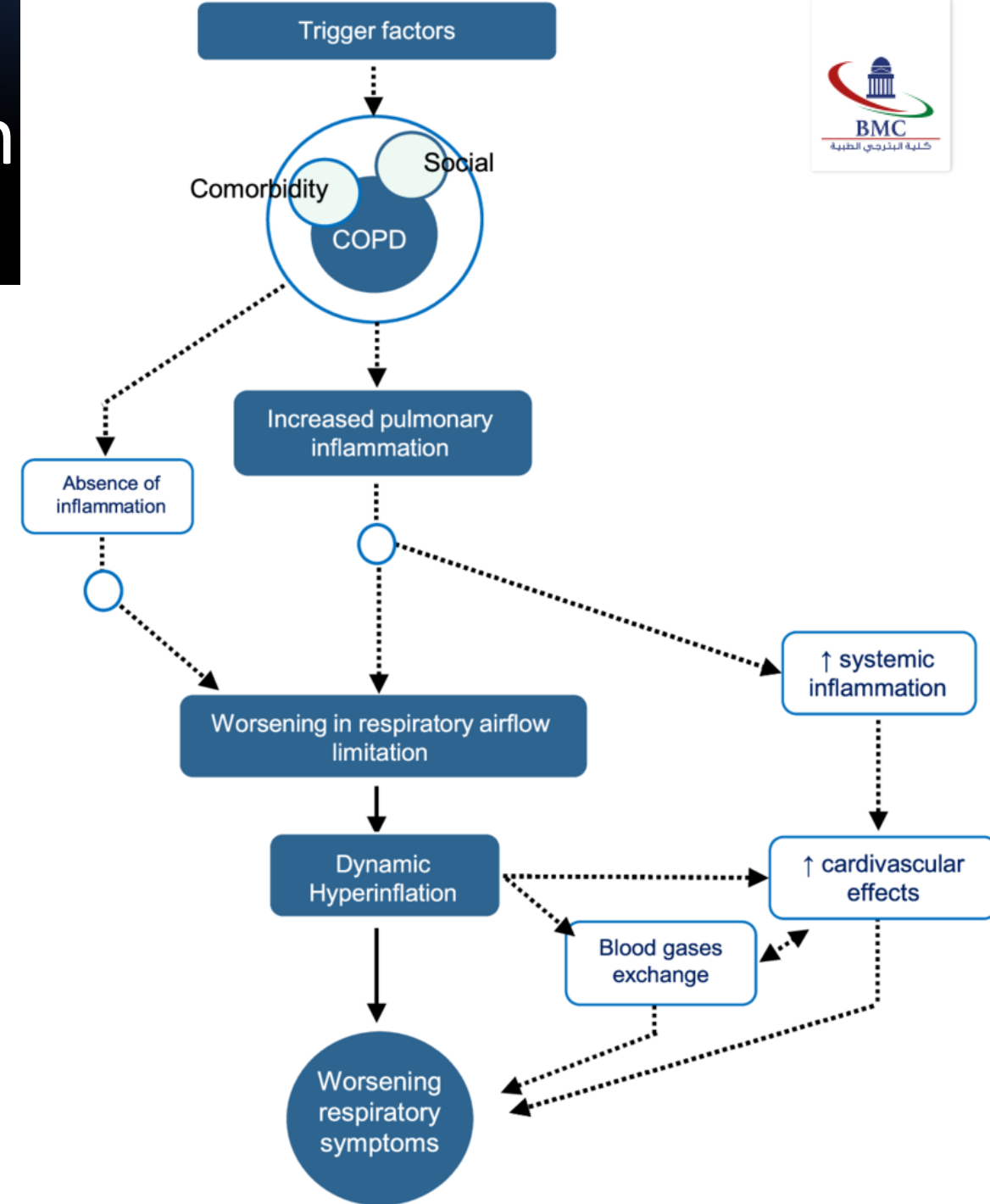
worsening dyspnea, ↑ cough, and a change in sputum volume or purulence

Etiology

Viral or bacterial infection (pneumonia)

Investigations

CXR, ABG, Flu swab, Sputum cultures



Acute COPD Exacerbation

Management:

Hospital admission in AECOPD

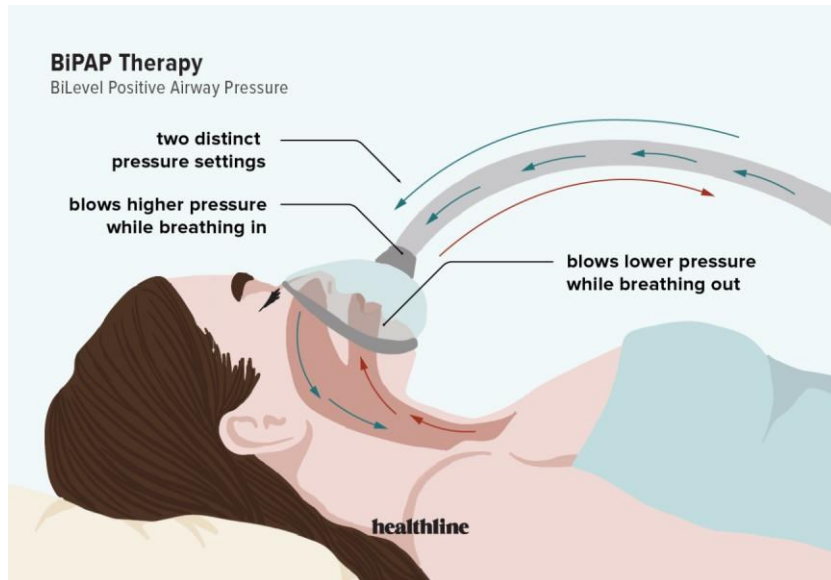
- Oxygen therapy as needed to maintain SpO_2 88–92%
- Start pharmacotherapy for AECOPD.
 - Start bronchodilators. (nebulized SBA, SAB/SAMA & ICS)
 - Consider systemic glucocorticoids.
 - Consider indications for antibiotic therapy in AECOPD. (profuse purulent sputum, require ventilation)
 - Provide adjunct pharmacological agents if clinically indicated, (e.g., diuretics for Pulmonary hypertension)
- Assess for signs of impending respiratory failure. (Cyanosis, Tachypnea) transfer to ICU
- Obtain serial ABGs.
- If patients have hypercapnia and/or persistent hypoxemia:
 - Titrate oxygen therapy to target SpO_2 .

Acute COPD Exacerbation

Consider a trial of NIPPV

Indication for NIPPV:

- Respiratory acidosis ($\text{PaCO}_2 \geq 45$ mm Hg and arterial $\text{pH} \leq 7.35$)
- Severe dyspnea with signs of increased work of breathing and/or respiratory muscle fatigue
- Persistent hypoxemia despite supplemental oxygen



Acute COPD Exacerbation

Consider indications for intubation in AECOPD.

- Contraindication of NIVPPV (facial trauma, Repeated vomiting, uncooperative patient)
- Confusion
- Persistent respiratory acidosis in spite NIVPPV
- Impending respiratory arrest
- Hemodynamic un stability



To Summarize

COPD
Chronic obstructive pulmonary disease

Epidemiology
Sex: 3:2 male/female ratio
Prevalence (US): ~6%
Third most common cause of death worldwide

Etiology/risk factors
Exogenous: tobacco smoking, air pollution, occupational exposures
Endogenous: α 1-Antitrypsin deficiency, antibody deficiency syndromes, premature birth

Diagnosis
Pulmonary function tests: 
Blood gas analysis
Imaging: chest x-ray, chest CT
 α 1-Antitrypsin if suspecting AATD

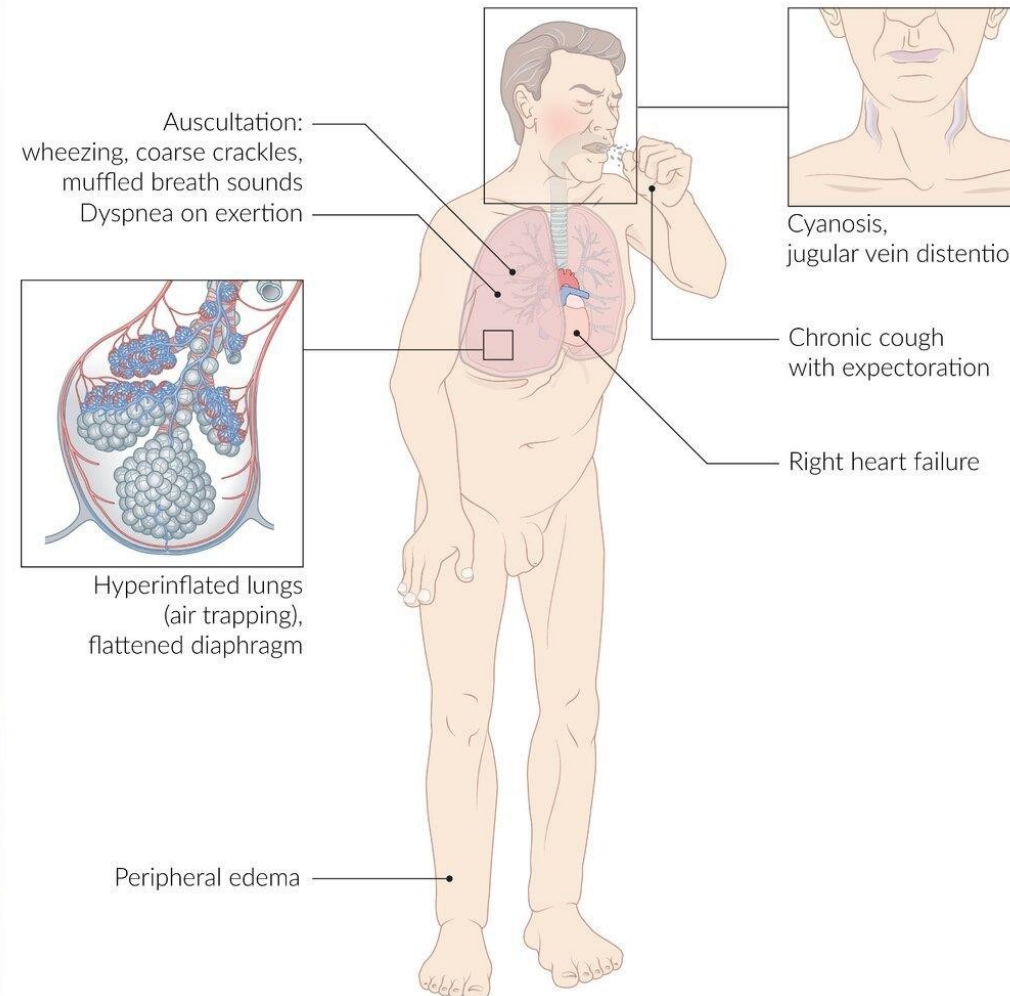
Complications
Acute exacerbations (AECOPD)
Chronic respiratory failure

Note
No curative therapy

GOLD classification

Classification based on airflow limitation in patients with $FEV_1/FVC < 70\%$

Category	Symptoms	FEV_1 % of the predicted value
GOLD 1 (Class I)	Mild	$\geq 80\%$
GOLD 2 (Class II)	Moderate	$50\% \leq FEV_1 < 80\%$
GOLD 3 (Class III)	Severe	$30\% \leq FEV_1 < 50\%$
GOLD 4 (Class IV)	Very severe	$< 30\%$



Case Scenario 1



A 68-year-old man comes to the physician for a routine health maintenance examination. He has had increasing shortness of breath and a productive cough for the past year. He can walk up a flight of stairs without stopping. He has hypertension and his only medication is amlodipine. He has a history of asthma during childhood. He used to work at a shipyard and retired 5 years ago. He smoked 1 pack of cigarettes daily for 30 years but quit 16 years ago. His pulse is 98/min, respirations are 25/min and blood pressure is 134/88 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 93%. Scattered expiratory wheezing and rhonchi are heard throughout both lung fields. Cardiac examination shows no murmurs, rubs, or gallops. Spirometry shows an $FEV_1:FVC$ ratio of 63% and an FEV_1 of 65% of predicted. His diffusing capacity for carbon monoxide (DL_{CO}) is 40% of predicted. Which of the following is the most likely cause of this patient's findings?

- A. Asthma
- B. Chronic bronchitis
- C. Congestive heart failure
- D. Pulmonary hypertension
- E. Bronchiectasis
- F. Emphysema

Case Scenario 2



A 65-year-old woman with chronic obstructive pulmonary disease comes to the emergency department with a 2-day history of worsening shortness of breath and cough. She has a mild productive cough for the past 2 years but she noticed that her sputum is more yellow than usual over the past 2 days. She has not had any recent fevers, chills, sore throat, or a runny nose. Her only medication is a salmeterol inhaler that she uses twice daily. Her temperature is 36.7°C (98.0°F), pulse is 104/min, respirations are 28/min, and blood pressure is 134/73 mm Hg. Pulse oximetry shows an oxygen saturation of 85% on room air. She appears uncomfortable and shows labored breathing. Auscultation of the lungs discloses rhonchi that disappear after coughing and diffuse bilateral wheezes. A chest x-ray shows mild hyperinflation and flattening of the diaphragm but no consolidation. She is started on supplemental oxygen via nasal cannula. Which of the following is the most appropriate initial pharmacotherapy?

- A. Prednisone and roflumilast
- B. Albuterol and montelukast
- C. Prednisone and salmeterol
- D. Albuterol and theophylline
- E. Prednisone and albuterol
- F. Prednisone and tiotropium

Questions?



Healthy Lung



Chronic Obstructive Pulmonary Disease (COPD)

References & further reading

- Step-up to medicine 5th edition
- Harrison's Principles of Internal Medicine 20th edition
- Amboss
- Cecil Essentials of Medicine 10th edition
- First Aid for the internal medicine board 4th edition

Thank You